Trigonometric Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Graphing tangent functions

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 4(5) + 9 = 29$.

2 [Foundational] • Target: Graphing tangent functions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

3 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 5(2) + 3 = 13$.

4 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

5 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

A data set recorded for key features of secant, cosecant, and cotangent graphs consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

6 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(3)$.

Final Answer: 18. Substitute $x = 3$: $f(3) = 5(3) + 3 = 18$.

7 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(4)$.

Final Answer: 23. Substitute $x = 4$: $f(4) = 5(4) + 3 = 23$.

9 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

10 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(5)$.

Final Answer: 28. Substitute $x = 5$: $f(5) = 5(5) + 3 = 28$.

11 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 [Intermediate] • Target: Graphing secant, cosecant, and cotangent functions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(2)$.

Final Answer: 16. Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

13 [Intermediate] • Target: Graphing secant, cosecant, and cotangent functions

A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

14 [Intermediate] • Target: Graphing secant, cosecant, and cotangent functions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(3)$.

Final Answer: 22. Substitute $x = 3$: $f(3) = 6(3) + 4 = 22$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphing secant, cosecant, and cotangent functions • Cross-Lesson: 3.01 Polynomial Functions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(4)$.

Final Answer: 28. Substitute $x = 4$: $f(4) = 6(4) + 4 = 28$.

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16 [Challenging] • Target: Graphing secant, cosecant, and cotangent functions • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(5)$.

Final Answer: 34. Substitute $x = 5$: $f(5) = 6(5) + 4 = 34$.

17 [Challenging] • Target: Graphing sine and cosine functions • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The cosine function is symmetric about the y-axis because it is:.

Final Answer: Even. Applying Trigonometric Functions principles, $\cos(-x) = \cos x$: even function. Therefore, the verified correct option is Even.

18 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 6.01 Radicals & Rational Exponents

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$? Show your complete algebraic work.

Final Answer: 24. We solve the problem in two key phases: identifying the polynomial $f(x)$ from its graphical features, and applying the Polynomial Remainder Theorem to determine $R(x)$. **Step 1: Determine $f(x)$** - Because the graph is tangent to the x-axis at $x = -2$, $(x + 2)$ is a factor with an even multiplicity of at least 2. - Because the graph crosses the x-axis at $x = 1$ and $x = 3$, $(x - 1)$ and $(x - 3)$ are factors with odd multiplicities of at least 1. - Since $f(x)$ is of degree 4, the.

19 [Challenging] • Target: Key features of secant, cosecant, and cotangent graphs • Cross-Lesson: 7.01 Sequences & Series

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have? Show your complete algebraic work.

Final Answer: 4. We solve the iterated equation $f(f(f(x))) = 0$ layer by layer from the outside in. **Step 1: Outer Layer** Let $u = (f \circ f)(x)$. The equation becomes: $f(u) = 0 \Rightarrow \frac{1}{2}u^2 - 2 = 0 \Rightarrow u^2 = 4 \Rightarrow u = -2 \quad \text{or} \quad u = 2$ **Step 2: Middle Layer** Now set $v = f(x)$ and solve $f(v) = u$ for each value of u: - Case A: $f(v) = -2$ $\frac{1}{2}v^2 - 2 = -2 \Rightarrow \frac{1}{2}v^2 = 0 \Rightarrow v = 0$ Since -2 is the minimum value (vertex) of t.

20 [Challenging] • Target: Graphing secant, cosecant, and cotangent functions • Cross-Lesson: 8.01 Conic Sections

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$? Show your complete algebraic work.

Final Answer: 4. The maximum value of a function refers to the greatest output value (or y-value) that the function attains across its domain. The given quadratic function is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = -1$, $h = 3$, and $k = 4$. 1. For any real number x, the squared term is non-negative: $(x - 3)^2 \geq 0$. 2. Multiplying by -1 reverses the inequality: $-(x - 3)^2 \leq 0$. 3. Adding 4 to both sides gives: $-(x - 3)^2 + 4 \leq 4$, which means $f(x) \leq 4$ for all x. When.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(2)$.

a. 11 ✓

b. 13

c. 4

d. 15

Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

22 [Foundational] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(5)$.

a. 17 ✓

b. 19

c. 10

d. 21

Substitute $x = 5$: $f(5) = 2(5) + 7 = 17$.

24 [Foundational] • Target: Graphing sine and cosine functions

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(2)$.

a. 14 ✓

b. 17

c. 6

d. 18

Substitute $x = 2$: $f(2) = 3(2) + 8 = 14$.

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25 [Foundational] • Target: Graphing sine and cosine functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Graphing sine and cosine functions

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(5)$.

a. 23 ✓

b. 26

c. 15

d. 27

Substitute $x = 5$: $f(5) = 3(5) + 8 = 23$.

27 [Foundational] • Target: Graphing tangent functions

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(2)$.

a. 17 ✓

b. 21

c. 8

d. 21

Substitute $x = 2$: $f(2) = 4(2) + 9 = 17$.

28 [Foundational] • Target: Graphing tangent functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

31 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(3)$.

a. 13 ✓

b. 15

c. 6

d. 17

Substitute $x = 3$: $f(3) = 2(3) + 7 = 13$.

32 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(4)$.

a. 15 ✓

b. 17

c. 8

d. 19

Substitute $x = 4$: $f(4) = 2(4) + 7 = 15$.

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33 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

34 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

35 [Intermediate] • Target: Graphing sine and cosine functions

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

36 [Intermediate] • Target: Graphing sine and cosine functions

A data set recorded for graphing sine and cosine functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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37 [Intermediate] • Target: Graphing sine and cosine functions

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(3)$.

a. 17 ✓

b. 20

c. 9

d. 21

Substitute $x = 3$: $f(3) = 3(3) + 8 = 17$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphing sine and cosine functions • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

39 [Challenging] • Target: Graphing sine and cosine functions • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

40 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 8.01 Conic Sections

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

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41 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 1.01 Functions & Their Graphs

A data set recorded for graphing tangent functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

42 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(3)$.

a. 21 ✓

b. 25

c. 12

d. 25

Substitute $x = 3$: $f(3) = 4(3) + 9 = 21$.

43 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 11.01 Probability & Statistics

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(4)$.

a. 25 ✓

b. 29

c. 16

d. 29

Substitute $x = 4$: $f(4) = 4(4) + 9 = 25$.

44 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

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45 [Challenging] • Target: Key features of secant, cosecant, and cotangent graphs • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for key features of secant, cosecant, and cotangent graphs consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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